

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: df = pd.read_csv("dsbda.csv")
print("First 5 rows:\n", df.head())
```

First 5 rows:

	StudentID	Age	Gender	Ethnicity	ParentalEducation	StudyTimeWeekly	\
0	1001	17	1	0	2	19.833723	
1	1002	18	0	0	1	15.408756	
2	1003	15	0	2	3	4.210570	
3	1004	17	1	0	3	10.028829	
4	1005	17	1	0	2	4.672495	

	Absences	Tutoring	ParentalSupport	Extracurricular	Sports	Music	\
0	7	1	2	0	0	1	
1	0	0	1	0	0	0	
2	26	0	2	0	0	0	
3	14	0	3	1	0	0	
4	17	1	3	0	0	0	

	Volunteering	GPA	GradeClass
0	0	2.929196	2.0
1	0	3.042915	1.0
2	0	0.112602	4.0
3	0	2.054218	3.0
4	0	1.288061	4.0

```
In [3]: print("\nMissing values:\n", df.isnull().sum())
print("\nStatistical Summary:\n", df.describe())
```

Missing values:

```

StudentID      0
Age            0
Gender         0
Ethnicity      0
ParentalEducation  0
StudyTimeWeekly  0
Absences       0
Tutoring       0
ParentalSupport  0
Extracurricular  0
Sports         0
Music          0
Volunteering   0
GPA            0
GradeClass     0
dtype: int64

```

Statistical Summary:

	StudentID	Age	Gender	Ethnicity	ParentalEducation \
count	2392.000000	2392.000000	2392.000000	2392.000000	2392.000000
mean	2196.500000	16.468645	0.510870	0.877508	1.746237
std	690.655244	1.123798	0.499986	1.028476	1.000411
min	1001.000000	15.000000	0.000000	0.000000	0.000000
25%	1598.750000	15.000000	0.000000	0.000000	1.000000
50%	2196.500000	16.000000	1.000000	0.000000	2.000000
75%	2794.250000	17.000000	1.000000	2.000000	2.000000
max	3392.000000	18.000000	1.000000	3.000000	4.000000

	StudyTimeWeekly	Absences	Tutoring	ParentalSupport \
count	2392.000000	2392.000000	2392.000000	2392.000000
mean	9.771992	14.541388	0.301421	2.122074
std	5.652774	8.467417	0.458971	1.122813
min	0.001057	0.000000	0.000000	0.000000
25%	5.043079	7.000000	0.000000	1.000000
50%	9.705363	15.000000	0.000000	2.000000
75%	14.408410	22.000000	1.000000	3.000000
max	19.978094	29.000000	1.000000	4.000000

	Extracurricular	Sports	Music	Volunteering	GPA \
count	2392.000000	2392.000000	2392.000000	2392.000000	2392.000000
mean	0.383361	0.303512	0.196906	0.157191	1.906186
std	0.486307	0.459870	0.397744	0.364057	0.915156
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	1.174803
50%	0.000000	0.000000	0.000000	0.000000	1.893393
75%	1.000000	1.000000	0.000000	0.000000	2.622216
max	1.000000	1.000000	1.000000	1.000000	4.000000

	GradeClass
count	2392.000000
mean	2.983696
std	1.233908
min	0.000000
25%	2.000000
50%	4.000000

```
75%      4.000000
max      4.000000
```

```
In [4]: print("\nRows:", df.shape[0])
        print("Columns:", df.shape[1])
        print("\nData Types:\n", df.dtypes)
```

```
Rows: 2392
Columns: 15
```

```
Data Types:
StudentID      int64
Age            int64
Gender         int64
Ethnicity      int64
ParentalEducation  int64
StudyTimeWeekly  float64
Absences       int64
Tutoring       int64
ParentalSupport  int64
Extracurricular  int64
Sports         int64
Music          int64
Volunteering   int64
GPA            float64
GradeClass     float64
dtype: object
```

```
In [5]: df['Gender'] = df['Gender'].astype('category')
        df['Sports'] = df['Sports'].astype('category')
        df['Music'] = df['Music'].astype('category')
        df['Tutoring'] = df['Tutoring'].astype('category')

        print("\nUpdated Data Types:\n", df.dtypes)
```

```
Updated Data Types:
StudentID      int64
Age            int64
Gender         category
Ethnicity      int64
ParentalEducation  int64
StudyTimeWeekly  float64
Absences       int64
Tutoring       category
ParentalSupport  int64
Extracurricular  int64
Sports         category
Music          category
Volunteering   int64
GPA            float64
GradeClass     float64
dtype: object
```

```
In [6]: df_encoded = pd.get_dummies(df, columns=[
        'Gender',
        'Sports',
        'Music',
```

```
'Tutoring']])
```

```
print("\nEncoded Data:\n", df_encoded.head())
```

Encoded Data:

	StudentID	Age	Ethnicity	ParentalEducation	StudyTimeWeekly	Absences	\
0	1001	17	0	2	19.833723	7	
1	1002	18	0	1	15.408756	0	
2	1003	15	2	3	4.210570	26	
3	1004	17	0	3	10.028829	14	
4	1005	17	0	2	4.672495	17	

	ParentalSupport	Extracurricular	Volunteering	GPA	GradeClass	\
0	2	0	0	2.929196	2.0	
1	1	0	0	3.042915	1.0	
2	2	0	0	0.112602	4.0	
3	3	1	0	2.054218	3.0	
4	3	0	0	1.288061	4.0	

	Gender_0	Gender_1	Sports_0	Sports_1	Music_0	Music_1	Tutoring_0	\
0	False	True	True	False	False	True	False	
1	True	False	True	False	True	False	True	
2	True	False	True	False	True	False	True	
3	False	True	True	False	True	False	True	
4	False	True	True	False	True	False	False	

	Tutoring_1
0	True
1	False
2	False
3	False
4	True

```
In [7]: print("\nFinal Shape:", df_encoded.shape)
```

Final Shape: (2392, 19)

```
In [ ]:
```